



Tarang Srivas
DOB: 22/02/2003
Roll No.: 2024PVL0107
M.Tech - VLSI Design
Indian Institute Of Technology, Jammu

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tarang321
in Tarang Srivas

EDUCATION

Degree/Certificate	Institute/Board	CGPA/Percentage	Year of Passing
M.Tech (VLSI Design)	Indian Institute of Technology, Jammu	8.35(Current)	2026
B.Tech (ECE(IoT))	Sir Padampat Singhania University,Udaipur	8.808	2024

EXPERIENCE

- Prileo Private Limited

Jan 2024 - May 2024

Embedded system Engineer

Bangalore

– Architected a complete IoT-based home automation solution from concept to deployment, integrating embedded firmware, hardware circuit design, and mobile app development, enabling real-time monitoring, control, and cloud connectivity for 20+ devices.

– Deployed FreeRTOS-based task scheduling on ESP32, boosting concurrent process efficiency by 35% and cutting sensor data latency by 40% .

– Directed end-to-end product life cycle — from system architecture and PCB design to 3D CAD modeling — delivering the prototype 2 weeks ahead of schedule and reducing costs by 15% via in-house fabrication.
- RoomsVitals

July 2023 - Oct 2023

Embedded System Engineer

Chandigarh

– Crafted a smart doorbell and wireless PLC for IoT-based home automation, achieving 99.5% device-to-cloud up-time in live deployments.

– Optimized communication protocols and device integration, improving system response time by 25% and ensuring seamless operation across 10+ devices.
- Nuclear Power Corporation of India Limited

May 2023 - July 2023

Vocational Trainee

Rawatbhata

– Configured and validated PLC-based control systems for industrial automation, reducing fault detection time by 20% and enhancing production line reliability.

– Documented and analyzed operational workflows and safety protocols in a nuclear power plant, strengthening knowledge of process automation and advanced control systems, supporting compliance with a 100% safety audit pass rate.

TECHNICAL SKILLS

- **Programming:** C, C++, DSA, Embedded C, Python*

• **Database:** ThinkSpeak, Firebase, AWS

• **Operating Systems:** Windows, Raspbian*, Linux, RTOS

• **Protocols:** UART, SPI, I2C, Wi-Fi, Bluetooth, MQTT

• **Tools:** Git, Easy EDA, Virtuoso, Vivado, Canva, Mission Planner

* Elementary proficiency

PROJECTS

- Surveillance optical Tethered Drone

Indian Army

Dr.Padmini Singh

4 Jammu and Kashmir Light Infantry

– Conceived and delivered a military-grade tethered UAV for continuous surveillance, leveraging optical communication to provide secure, high-bandwidth data transfer (up to 1.2 Gbps) with zero signal interference during 12-hour test operations.

– Devised a custom tethered power and communication system, extending operational endurance by 400% compared to battery-based UAVs.

– Constructed all hardware in-house, including PCB design, embedded firmware programming, and optical transceiver integration, achieving sub-200 ms control latency and stable HD video streaming in adverse weather conditions.
- Guardiano

July 2024 - Nov 2024

Course Project / Dr.Ankit Dubey, Assistant Professor, Dept. of Electrical, IIT Jammu

– Created a humanoid companion robot with embedded facial recognition for child safety, achieving 97% identification accuracy in real-time tests.

– Applied FreeRTOS-based task scheduling on ESP32 to handle concurrent processes, improving task execution efficiency by 40%.

- Incorporated Raspberry Pi for AI-driven processing and Firebase Database for real-time remote monitoring, ensuring sub-200 ms response time and 99.8% data synchronization reliability.
- **Retro Fitting Smart IoT-Enabled Switchboard**
 - Produced a retrofit IoT-enabled switchboard to modernize legacy electrical infrastructure without rewiring, reducing upgrade cost by 60%.
 - Configured FreeRTOS on ESP32 to manage concurrent tasks, improving multi-device control responsiveness by 35%.
 - Formulated priority-based task scheduling to guarantee instant (sub-100 ms) manual switch response while sustaining uninterrupted remote connectivity.
 - Facilitated real-time monitoring and control via a mobile app and cloud dashboard, ensuring 99.5% up-time and automatic recovery from faults within 5 seconds.
- **12 Channel 1.5 km Ranged RF receiver and transmitter Board Fully handmade PCB**
 - Authored and assembled a 12-channel RF receiver and transmitter board with a verified 1.5 km range and 98% packet delivery rate in open-field testing.
 - Fabricated a single-layer PCB through manual etching and drilling, achieving a 95% first-pass success rate and cutting prototyping time by 30% compared to outsourced manufacturing.
- **Other Projects** 📁

KEY COURSES TAKEN

- Advanced Digital System Design, Introduction to IoT, Machine Learning, Deep Learning, Robotics and Automation System

ACHIEVEMENTS

- GATE ECE 2024 – Secured All India Rank 6160 *2024*
- NCC ‘C’ Certificate ‘A’ Grade; National Trial Trek Camp *2023–2022*
- International Conventions – Spicmacay, VNIT Nagpur *2023*
- Prototype & Concept Presentation Winner; IEEE Ideation Gold & Bronze Medalist *2023–2022*
- Intel IRIS Fair, Indian Science Congress; AEES Prize Recipient *2019–2018*
- National Science Exhibition – 1st Place & 2nd Runner-up (Multiple Years) *2016–2018*